

Joseph J. Webb

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education

- Doctor of Philosophy in Mathematics

Sep 2025—present

University of Oxford, Worcester College

Oxford, UK

Supported by the EPSRC [EP/Z534870/1], through an iCASE studentship in collaboration with NAG Ltd.
- Master of Science in Mathematical Modelling and Scientific Computing

Sep 2024—Sep 2025

University of Oxford, Worcester College

Oxford, UK

The **Worcester College Mathematical Sciences Scholar**, recipient of a Mathematical Institute bursary for academic excellence, and winner of **The Kathryn Gillow Prize for the Best Dissertation**.
- Bachelor of Science in Mathematics and Physics

Sep 2021—Aug 2024

University of Warwick

Coventry, UK

Founder of **The Poincaré Project** which produced a yearly Mathematics, Physics and Computer Science magazine.

experience

- Self-Employed

Jan 2017—Aug 2021

Electrical Inspector

London, UK

Contracted electrical inspecting and testing of financial institutions and data centres in Central London.
- UK Government, Department of Digital, Culture, Media & Sport

Jun 2017—Jun 2017

Information Technology Support Technician

London, UK

Provided 1st- and 2nd-line support and produced technical documentation.
- The Economist

Feb 2020—Sep 2020

Information Technology Operations Analyst

London, UK

Created scheduled scripts to automate server tasks, and built software to optimise resolution of issues.
- Selfridges & Co.

Feb 2019—Oct 2019

Computer Services Analyst

London, UK

Improved our rollout speed by developing a GUI application to perform device management.

publications

- Joseph J. Webb, Sadok Jerad, Coralia Cartis, **An Optimisation Framework for the Well-Conditioned Training of Physics-Informed Neural Networks**, *arXiv preprint*.

doi.org/10.48550/arXiv.2607.02194 ↗
- Joseph J. Webb, Richard Beanland, Rudolf Römer, **Large-Angle Convergent-Beam Electron Diffraction Patterns via Conditional Generative Adversarial Networks**, *Ultramicroscopy*.

doi.org/10.1016/j.ultramic.2025.114198 ↗

articles & projects

- Article // **How to Win Games with Quantum Strategies**. *Poincaré Magazine* **2**, 23-27 (2024). ↗
- Article // **A Walk in the Quantum: Reinventing a World-Changing Algorithm**. *Poincaré Magazine* **1**, 29-36 (2022). ↗
- Code // **Physics-Informed Neural Networks**. A JAX framework for solving PDEs with physics-informed neural networks.

github.com/wephy/physics-informed-neural-networks ↗
- Code // **Quantum Walk**. Emulator for quantum walks on directed graphs to speed up the classical PageRank algorithm.

github.com/wephy/quantum-walk ↗